EMCH 368 Mechatronics
Fall 2026

Self-study

Practice Set 1 Solution

100 points

Answer key. The questions and answer boxes match the practice sheet. Worked calculations follow each group of boxes.

Problem 1. (10 pt) True or False

Each statement is worth 1 point. Assume ideal components where stated.

1. A negative-feedback error signal equals the reference value minus the measured value.
2. An error specified as a percentage of full scale changes whenever the current reading changes.
3. The span of a sensor is its upper input limit minus its lower input limit.
4. An ideal op-amp draws zero current at its input terminals.
5. A voltage follower has a voltage gain of zero.
6. After one time constant, a first-order sensor has completed about 36.8% of a step change.
7. A balanced Wheatstone bridge has equal midpoint voltages.
8. PWM duty cycle is the fraction of each period for which the signal is high.
9. Increasing an ADC from 8 bits to 10 bits halves its voltage step size for the same input span.
10. A two-input XOR gate outputs 1 when both inputs are 1.

Number	T/F	Number	T/F
1	**T**	6	**F**
2	**F**	7	**T**
3	**T**	8	**T**
4	**T**	9	**F**
5	**F**	10	**F**

Problem 1. Answer explanations

1. **T.** The control-error convention is $e = r - y_m$.

2. **F.** A full-scale specification uses the fixed measurement span. For example, a 0.2% error over a 500 kPa span is 1 kPa at every reading.

3. **T.** $S = x_{\max} - x_{\min}$. The range is the interval between those endpoints.

4. **T.** Infinite ideal input impedance gives $I_+ = I_- = 0$.

5. **F.** A follower has $V_o = V_i$, so its voltage gain is 1.

6. **F.** The completed fraction is $1 - e^{-1} = 0.6321$, or 63.2%. The remaining fraction is 36.8%.

7. **T.** Equal midpoint potentials give zero differential output voltage.

8. **T.** $D = t_{\text{on}}/T$.

9. **F.** Two extra bits increase the number of levels by $2^2 = 4$, so the step size becomes one quarter of its former value.

10. **F.** XOR outputs 1 when its inputs differ; its output for two 1s is 0.

Problem 2. (10 pt) Measurement error and calibration

A pressure sensor has an input range of -50 to 450 kPa and an output range of 1 to 5 V. Treat the following tests as independent.

(a) (2 pt) At a true pressure of 200 kPa, its display reads 196 kPa. Find the input span and signed measurement error.

(b) (2 pt) Nonlinearity and hysteresis limits are $\pm 0.20\%$ and $\pm 0.10\%$ of full input span. Find the combined worst-case error bound in kPa.

(c) (3 pt) In a separate calibration test, the output at 200 kPa is 3.04 V. Use a straight line through the range endpoints to find sensitivity, predicted output, and the output deviation as a percentage of full output span.

(d) (3 pt) Repeated outputs at one fixed input are 3.01, 3.03, and 3.00 V. Find percentage repeatability error. Also find the control error if a controller requests 250 kPa and receives a 245 kPa measurement.

(a) Input span; signed measurement error
$S = 500\ \text{kPa}; \qquad e_m = -4.00\ \text{kPa}$

(b) Combined worst-case error bound
$e \in [-1.50, +1.50]\ \text{kPa}$

(c) Sensitivity; predicted output; percentage deviation
$K = 0.00800\ \text{V/kPa}; \quad V_{\text{line}} = 3.00\ \text{V}$ $e_{\text{NL}} = +0.0400\ \text{V} = +1.00\%$ of span

(d) Repeatability error; control error
$p_{\text{rep}} = 0.750\%; \qquad e = +5.00\ \text{kPa}$

Solution (Problem 2): The input span is $450 - (-50) = 500$ kPa and the measurement error is $196 - 200 = -4$ kPa. The error bounds add:

$$E_{\max} = (0.0020 + 0.0010)(500) = 1.50 \,\text{kPa}.$$

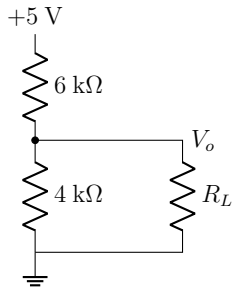
The calibration slope is

$$K = \frac{5 - 1}{450 - (-50)} = 0.00800 \,\text{V/kPa}.$$

At 200 kPa, $V_{\text{line}} = 1 + 0.00800[200 - (-50)] = 3.00$ V. Thus the signed output deviation is $3.04 - 3.00 = 0.04$ V, or $100(0.04)/4 = 1.00\%$ of output span. The repeatability error is $100(3.03 - 3.00)/4 = 0.750\%$. Finally, the control-error convention gives $250 - 245 = +5$ kPa.

Problem 3. (10 pt) Potentiometers, strain gauges and encoders

(a) (3 pt) A uniform $10\ \mathrm{k}\Omega$ potentiometer is supplied by 5 V. The slider is 40% of the way from the grounded end. Find the unloaded output and the output when a $10\ \mathrm{k}\Omega$ load is connected from slider to ground.



(b) (3 pt) A strain gauge has $R_0 = 350\ \Omega$ and gauge factor 2.00. Its resistance increases to $350.42\ \Omega$. Find strain and microstrain. Neglect temperature changes.

(c) (4 pt) An encoder generates 360 pulses/revolution. A tracking wheel of radius 30 mm turns without slipping and produces 480 pulses in 0.800 s. Find angular resolution, linear travel, and average shaft speed in rpm. Count one pulse per slot.

(a) Unloaded output; loaded output
$V_{o,\mathrm{unloaded}} = 2.00\ \mathrm{V}; \quad V_{o,\mathrm{loaded}} = 1.613\ \mathrm{V}$

(b) Strain; microstrain
$\varepsilon = 6.00 \times 10^{-4}; \qquad 600\ \mu\varepsilon$

(c) Angular resolution; travel; average speed
$\Delta\theta = 1.00^\circ/\mathrm{pulse}; \quad x = 251.3\ \mathrm{mm}$ $n = 100\ \mathrm{rpm}$

Solution (Problem 3): The unloaded divider gives $V_o = 5(4/10) = 2.00$ V. With the load, the lower resistance is

$$R_p = 4 \parallel 10 = \frac{4(10)}{4+10} = 2.85714\ \text{k}\Omega.$$

Then $V_o = 5(2.85714)/(6 + 2.85714) = 1.61290$ V. For the gauge,

$$\varepsilon = \frac{\Delta R}{G_F R_0} = \frac{350.42 - 350}{2(350)} = 0.000600.$$

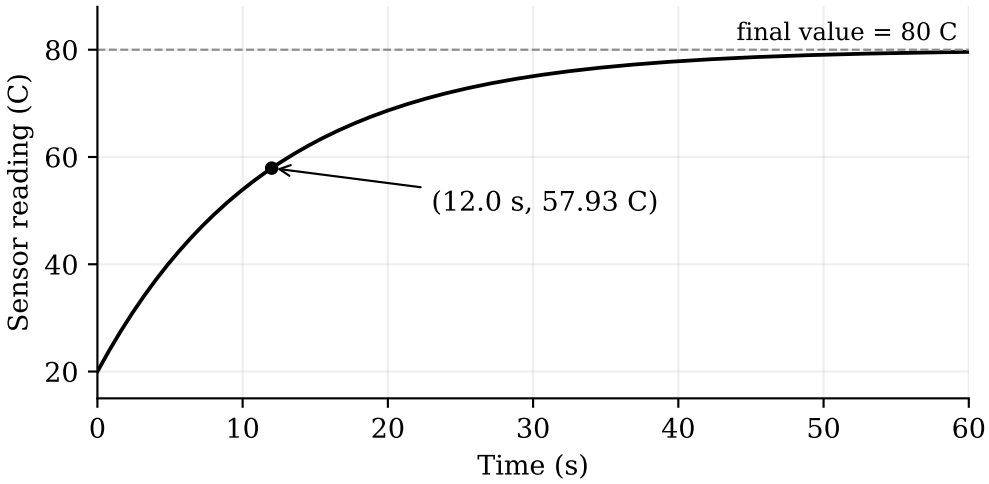
Multiplication by 10^6 gives $600\,\mu\varepsilon$. For the encoder,

$$\Delta\theta = \frac{360^\circ}{360} = 1.00^\circ, \quad x = \frac{480}{360}(2\pi \cdot 30) = 251.327\ \text{mm}.$$

The pulse frequency is $480/0.800 = 600$ pulses/s, so shaft speed is $60(600)/360 = 100$ rpm.

Problem 4. (10 pt) Sensor response and time constant

A temperature sensor initially reads $20\,^{\circ}\mathrm{C}$. At $t = 0$ it is moved into a bath held at $80\,^{\circ}\mathrm{C}$. Assume a first-order response with no dead time. A measured point is marked below.



(a) (3 pt) Determine the time constant τ.

(b) (3 pt) Write the complete sensor response $T(t)$ for $t \geq 0$.

(c) (2 pt) Calculate the reading after 18.0 s.

(d) (2 pt) Find the time to complete 95% of the total temperature change.

(a) Time constant
$\tau = 12.0\ \mathrm{s}$

(b) Sensor response
$T(t) = 80 - 60e^{-t/12} \quad [^{\circ}\mathrm{C}], \quad t$ in s

(c) Reading after 18.0 s
$T(18) = 66.6\,^{\circ}\mathrm{C}$

(d) 95% response time
$t_{95} = 35.9\ \mathrm{s}$

Solution (Problem 4): The total step change is $80 - 20 = 60\,^{\circ}\mathrm{C}$. At the marked point the completed fraction is $(57.93 - 20)/60 \approx 0.6322$. A first-order response completes about 63.2% at one time constant, so $\tau \approx 12.0$ s. The response is

$$T(t) = T_{\infty} + (T_0 - T_{\infty})e^{-t/\tau} = 80 - 60e^{-t/12}.$$

Therefore $T(18) = 80 - 60e^{-1.5} = 66.6122\,^{\circ}\mathrm{C}$. For 95% completion, $1 - e^{-t/12} = 0.95$, giving

$$t_{95} = -12\ln(0.05) = 35.9488\ \mathrm{s}.$$

The 95% target is $20 + 0.95(60) = 77\,^{\circ}\mathrm{C}$, not 95% of the final temperature measured from zero.

Problem 5. (10 pt) Temperature, liquid level and flow

(a) (2 pt) A platinum RTD has $R_0 = 100\ \Omega$ at $0\,^\circ\mathrm{C}$ and $\alpha = 0.00385/^\circ\mathrm{C}$. Find temperature when its resistance is $119.25\ \Omega$, using the linear model.

(b) (2 pt) A thermocouple reads 5.20 mV with its cold junction at $25\,^\circ\mathrm{C}$. Use the supplied calibration data to find the equivalent zero-reference EMF and hot-junction temperature.

Temperature (°C)	EMF relative to $0\,^\circ\mathrm{C}$ (mV)
0	0.00
25	1.00
150	6.20

(c) (3 pt) A cylindrical tank has diameter 0.600 m and holds a liquid of density $900\ \mathrm{kg/m^3}$ to a depth of 1.50 m. The gas above the liquid is at 20.0 kPa gauge pressure. Find liquid weight and bottom gauge pressure. Use $g = 9.81\ \mathrm{m/s^2}$.

(d) (3 pt) Water flows steadily through a horizontal constriction. The upstream and throat areas are $A_1 = 0.00400\ \mathrm{m^2}$ and $A_2 = 0.00100\ \mathrm{m^2}$. The pressure drop is 8.00 kPa. Find ideal volumetric flow in L/s. Take $\rho = 1000\ \mathrm{kg/m^3}$ and neglect losses.

(a) RTD temperature
$T = 50.0\,^\circ\mathrm{C}$

(b) Corrected EMF; hot-junction temperature
$E(T_h, 0) = 6.20\ \mathrm{mV}; \qquad T_h = 150\,^\circ\mathrm{C}$

(c) Liquid weight; bottom gauge pressure
$W_\ell = 3.745\ \mathrm{kN}; \qquad p_{b,g} = 33.24\ \mathrm{kPa}$

(d) Ideal volumetric flow
$Q = 4.131\ \mathrm{L/s}$

Solution (Problem 5): For the RTD, solve $R = R_0(1 + \alpha T)$:

$$T = \frac{119.25/100 - 1}{0.00385} = 50.0\,^\circ\mathrm{C}.$$

Thermocouple compensation adds the cold-junction EMF:

$$E(T_h, 0) = 5.20 + 1.00 = 6.20\ \mathrm{mV}.$$

The supplied table assigns this EMF to $150\,^\circ\mathrm{C}$. The tank area is $A = \pi(0.600)^2/4 = 0.282743\ \mathrm{m}^2$. Thus

$$W_\ell = \rho g A h = 900(9.81)(0.282743)(1.50) = 3744.51\ \mathrm{N}.$$

The liquid adds $\rho g h = 13243.5$ Pa $= 13.2435$ kPa, so bottom gauge pressure is $20.0 + 13.2435 = 33.2435$ kPa. For the constriction, continuity and Bernoulli give

$$Q = \frac{A_2}{\sqrt{1 - (A_2/A_1)^2}} \sqrt{\frac{2\Delta p}{\rho}}.$$

Substituting $\Delta p = 8000$ Pa,

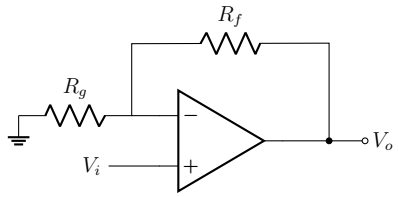
$$Q = \frac{0.00100}{\sqrt{1 - (0.25)^2}} \sqrt{\frac{2(8000)}{1000}} = 0.00413118\ \mathrm{m}^3/\mathrm{s}.$$

Since $1\ \mathrm{m}^3 = 1000$ L, $Q = 4.13118$ L/s.

Problem 6. (10 pt) Operational-amplifier circuits

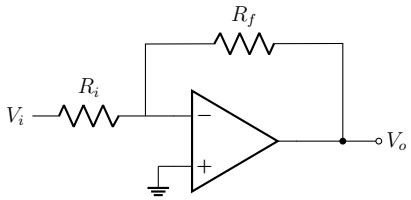
Assume ideal op-amps and adequate supply voltage to avoid saturation, except for the comparator in part (e).

(a) (2 pt) For the non-inverting circuit below, $V_i = 0.120$ V, $R_g = 10.0$ kΩ, and $R_f = 150$ kΩ. Find voltage gain and output.



(a) Gain; output voltage
$G = 16.0$; $\quad V_o = 1.92$ V

(b) (2 pt) For the inverting circuit, $V_i = -0.250$ V, $R_i = 10.0$ kΩ, and $R_f = 40.0$ kΩ. Find voltage gain and output.



(b) Gain; output voltage
$G = -4.00$; $\quad V_o = +1.00$ V

Solution (Problem 6): For a non-inverting amplifier,

$$G = 1 + \frac{R_f}{R_g} = 1 + \frac{150}{10} = 16, \qquad V_o = 16(0.120) = 1.92 \text{ V}.$$

For an inverting amplifier,

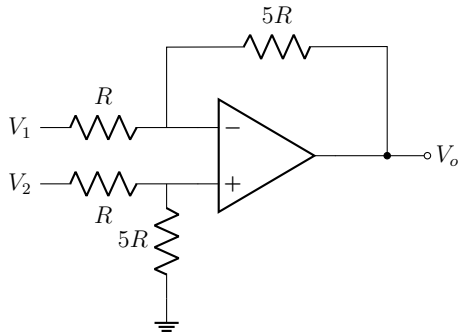
$$G = -\frac{R_f}{R_i} = -\frac{40}{10} = -4, \qquad V_o = (-4)(-0.250) = +1.00 \text{ V}.$$

Problem 6 (continued). Operational-amplifier circuits

(c) (2 pt) An inverting summing amplifier has its non-inverting input grounded. Input $V_A = 0.200$ V connects to the summing node through $R_A = 10.0\ \text{k}\Omega$, and $V_B = 0.600$ V connects through $R_B = 20.0\ \text{k}\Omega$. The feedback resistor is $R_f = 50.0\ \text{k}\Omega$. Find output voltage.

(c) Summing-amplifier output
$V_o = -2.50$ V

(d) (2 pt) In the difference amplifier below, both resistors labeled R are $10.0\ \text{k}\Omega$, and both labeled $5R$ are $50.0\ \text{k}\Omega$. Let $V_1 = 0.800$ V and $V_2 = 0.950$ V. Find output voltage.



(d) Difference-amplifier output
$V_o = +0.750$ V

(e) (2 pt) A comparator saturates at $+5.00$ V or -5.00 V. Its non-inverting input is 1.20 V and its inverting input is 1.50 V. Find output voltage.

(e) Comparator output
$V_o = -5.00$ V

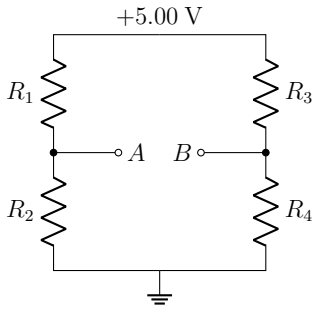
Solution (Problem 6): The summing amplifier weights each input by its own resistor:

$$V_o = -\left(\frac{50}{10}(0.200) + \frac{50}{20}(0.600)\right) = -2.50\ \text{V}.$$

The matched difference-amplifier ratio is 5, so $V_o = 5(V_2 - V_1) = 5(0.950 - 0.800) = 0.750$ V. For the comparator, $V_+ - V_- = 1.20 - 1.50 = -0.30$ V, so the output goes to the negative saturation level, -5.00 V.

Problem 7. (10 pt) Strain-gauge bridge and amplification

The bridge is supplied by an ideal 5.00 V source. Each arm is initially $120\ \Omega$. Only R_1 is an active strain gauge; its gauge factor is 2.10. It experiences tensile strain $800\,\mu\varepsilon$. Define $V_o = V_B - V_A$. Neglect lead resistance, temperature effects, and measurement loading.



(a) (2 pt) State the balance condition in terms of R_1, R_2, R_3, R_4. Find the initial bridge output.

(b) (2 pt) Find the change in R_1 and its new resistance.

(c) (3 pt) Calculate V_A, V_B, and the exact bridge output after strain.

(d) (3 pt) An ideal differential amplifier has $V_{\text{amp}} = 200V_o$. Find its output voltage after strain.

(a) Balance condition; initial output
$R_1R_4 = R_2R_3; \qquad V_{o,0} = 0\ \text{V}$

(b) Resistance change; strained resistance
$\Delta R_1 = 0.2016\ \Omega; \qquad R_1' = 120.2016\ \Omega$

(c) Midpoint voltages; bridge output
$V_A = 2.497902\ \text{V}; \quad V_B = 2.500000\ \text{V}$ $V_o = +2.09824\ \text{mV}$

(d) Amplifier output
$V_{\text{amp}} = +0.41965\ \text{V}$

Solution (Problem 7): A balanced bridge has $V_A = V_B$, so

$$\frac{R_2}{R_1 + R_2} = \frac{R_4}{R_3 + R_4} \quad \Rightarrow \quad R_1 R_4 = R_2 R_3.$$

The initially equal arms satisfy this condition. Convert microstrain before using the gauge factor:

$$\Delta R_1 = G_F \varepsilon R_0 = 2.10(800 \times 10^{-6})(120) = 0.2016\ \Omega.$$

After strain, the left upper arm is $120.2016\ \Omega$, giving

$$V_A = 5\frac{120}{120.2016 + 120} = 2.49790176\ \text{V}, \qquad V_B = 2.50\ \text{V}.$$

Then $V_o = V_B - V_A = 0.00209823748$ V, and

$$V_{\text{amp}} = 200(0.00209823748) = 0.4196475\ \text{V}.$$

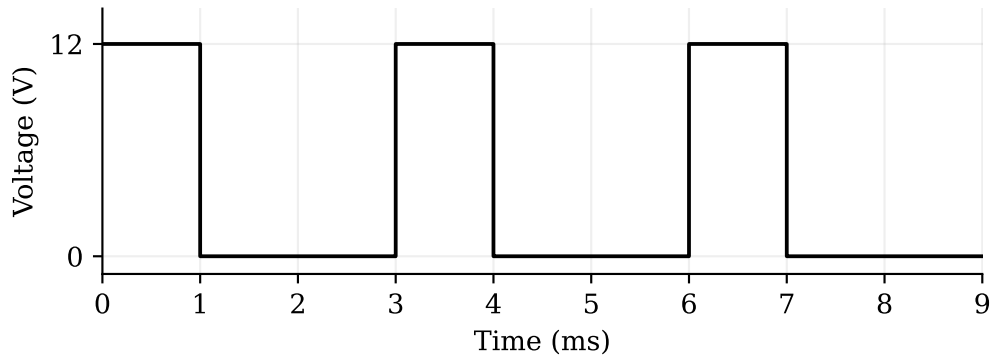
As a small-strain check, $V_o \approx (V_s/4) G_F \varepsilon = 2.100$ mV, close to the exact result. Increasing the upper-left resistance lowers V_A, which explains the positive sign of $V_B - V_A$.

Problem 8. (10 pt) Filtering, PWM and power transfer

(a) (4 pt) A first-order RC low-pass filter has $R = 10.0\ \text{k}\Omega$ and $C = 0.100\ \mu\text{F}$. Find its cutoff frequency. If the reference passband output amplitude is 2.00 V, find the output amplitude at cutoff and the attenuation relative to passband in dB. Assume negligible loading.

(a) Cutoff frequency; cutoff amplitude; attenuation
$f_c = 159.2\ \text{Hz}; \quad \lvert V_o(f_c)\rvert = 1.414\ \text{V}$ $A_{\text{dB}} = 3.010\ \text{dB}$

(b) (3 pt) The PWM signal below switches between 0 and 12.0 V. Determine period, pulse frequency, duty cycle, and average voltage.



(b) Period; frequency; duty cycle; average voltage
$T = 3.00\ \text{ms}; \quad f = 333.3\ \text{Hz}$ $D = 33.3\%; \quad V_{\text{avg}} = 4.00\ \text{V}$

(c) (3 pt) A DC source has EMF 12.0 V and internal resistance $6.00\ \Omega$. Find the load resistance for maximum power transfer and the maximum load power.

(c) Optimal load; maximum load power
$R_L = 6.00\ \Omega; \quad P_{L,\max} = 6.00\ \text{W}$

Solution (Problem 8): Convert $R = 10^4\ \Omega$ and $C = 10^{-7}$ F. Then

$$f_c = \frac{1}{2\pi RC} = 159.1549\ \text{Hz}, \quad |V_o(f_c)| = \frac{2}{\sqrt{2}} = 1.41421\ \text{V}.$$

The cutoff attenuation is $20\log_{10}(2/1.41421) = 3.0103$ dB. From the waveform, $t_{\text{on}} = 1$ ms and $T = 3$ ms, so

$$f = \frac{1}{0.003} = 333.333\ \text{Hz}, \quad D = \frac{1}{3}, \quad V_{\text{avg}} = 12\left(\frac{1}{3}\right) = 4\ \text{V}.$$

Maximum DC power occurs at $R_L = r = 6\ \Omega$. The current is $I = 12/(6+6) = 1$ A and the load power is $I^2 R_L = 6$ W. Equivalently, $P_{\max} = E^2/(4r) = 144/24 = 6$ W.

Problem 9. (10 pt) Conversion, sampling and binary data

(a) (2 pt) An ideal 12-bit ADC has an input range of 0–5.00 V. Find the voltage step size and maximum quantization-error magnitude for rounding to the nearest level. A temperature sensor has sensitivity $20.0\ \mathrm{mV/^\circ C}$; what temperature increment corresponds to one ADC step? Assume no overload.

(a) LSB size; maximum error; temperature increment
$q = 1.22070\ \mathrm{mV}; \quad \lvert e_q\rvert_{\max} = 0.61035\ \mathrm{mV}$ $\Delta T = 0.061035\,^\circ\mathrm{C}$

(b) (2 pt) Keeping the 0–5.00 V input range, how many bits are required so that the voltage step is no larger than 0.500 mV?

(b) Minimum number of bits
$n_{\min} = 14\ \mathrm{bits}$

(c) (2 pt) An ideal 8-bit DAC outputs exactly 0 V at code 0 and exactly 10.0 V at code 255. Find the output increment per code and the output at decimal code 153. Use these specified endpoints.

(c) DAC increment; output at code 153
$q_D = 39.2157\ \mathrm{mV/code}; \quad V_o = 6.00\ \mathrm{V}$

Solution (Problem 9): A 12-bit converter has $2^{12} = 4096$ codes. Therefore

$$q = \frac{5}{4096} = 0.001220703125\ \mathrm{V}, \qquad \lvert e_q\rvert_{\max} = \frac{q}{2}.$$

The physical-temperature increment is

$$\Delta T = \frac{q}{0.0200} = 0.0610352\,^\circ\mathrm{C}.$$

For the new resolution requirement,

$$n \geq \log_2\left(\frac{5}{0.000500}\right) = \log_2(10000) = 13.2877.$$

Round upward to 14 bits. A 13-bit step is 0.61035 mV and fails the requirement; a 14-bit step is 0.30518 mV and meets it. The DAC has 256 codes but 255 equal intervals between its specified endpoint voltages. Thus $q_D = 10/255 = 0.0392157$ V and $V_o = 153(10/255) = 6.00$ V.

Problem 9 (continued). Sampling and binary data

(d) (2 pt) A signal is band-limited to 350 Hz and is sampled at 1000 samples/s. Find the Nyquist frequency, state whether this sampling rate meets the ideal sampling criterion, and give a practical rate based on 10 samples per cycle of the highest frequency. State which filter type is placed before an ADC to suppress out-of-band high-frequency noise.

(d) Nyquist frequency; criterion; practical rate; filter type
$f_N = 500$ Hz; $\quad 1000 > 2(350) = 700$ Hz Ideal criterion: satisfied. Practical rate: 3500 samples/s; anti-aliasing low-pass filter.

(e) (2 pt) Convert decimal 173 to an 8-bit binary word. Find the even-parity bit to append. The parity bit is not included in the eight data bits.

(e) Eight-bit binary word; even-parity bit
$173_{10} = 10101101_2$; $\qquad p_{\text{even}} = 1$

Solution (Problem 9): The Nyquist frequency is half the sampling rate:

$$f_N = \frac{1000}{2} = 500 \text{ Hz}.$$

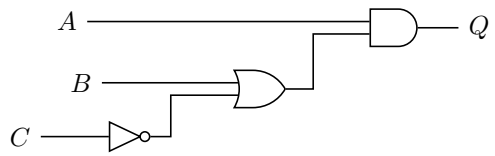
The 350 Hz signal is below this boundary. A practical 10-times rate is $10(350) = 3500$ samples/s. A low-pass anti-aliasing filter suppresses higher-frequency components before sampling; it must preserve the desired signal band and sufficiently attenuate components that would alias. For the binary conversion,

$$173 = 128 + 32 + 8 + 4 + 1.$$

Using place values $128, 64, 32, 16, 8, 4, 2, 1$ gives $1, 0, 1, 0, 1, 1, 0, 1$. There are five 1s, so the parity bit must be 1 to bring the total to six, an even number.

Problem 10. (10 pt) Digital logic and programming

(a) (3 pt) Write the Boolean expression for the circuit below in terms of inputs A, B, C.



(a) Boolean expression
$Q = A(B + \overline{C})$

(b) (3 pt) Complete the truth table. The plus sign in a Boolean expression means OR.

A	B	C	Q
0	0	0	**0**
0	0	1	**0**
0	1	0	**0**
0	1	1	**0**
1	0	0	**1**
1	0	1	**0**
1	1	0	**1**
1	1	1	**1**

Solution (Problem 10): The first gate complements C. The OR gate gives $B+\overline{C}$, and the final AND gate multiplies that Boolean result by A. Thus $Q = A(B+\overline{C})$. When $A = 0$, the output is 0 for every B, C pair. When $A = 1$, the output is 0 only for $B = 0, C = 1$; in the other three cases the OR gate outputs 1.

Problem 10 (continued). Programming calculations

(c) (2 pt) The user enters $n = 6$. Determine the displayed value of `sum`.

```
Get n
i = 0
sum = 0
While i < n
  sum = sum + 2*i
  i = i + 1
End while
Display sum
```

(c) Displayed sum
`sum` $= 30$

(d) (2 pt) A rectangular lot is 24.0 m by 18.0 m. A rectangular house occupying 12.0 m by 8.00 m lies within it. All remaining area is grass. A mower cuts $30.0\ \mathrm{m}^2/\mathrm{min}$. Write an expression for the mowing time and calculate it.

(d) Mowing-time expression and value
$t = \dfrac{L_{\mathrm{lot}}W_{\mathrm{lot}} - L_{\mathrm{house}}W_{\mathrm{house}}}{r_{\mathrm{mow}}}$ $t = 11.2\ \mathrm{min}$

Solution (Problem 10): The loop runs for $i = 0, 1, 2, 3, 4, 5$; the condition $i < 6$ excludes 6. It accumulates

$$0 + 2 + 4 + 6 + 8 + 10 = 30.$$

Equivalently, $\sum_{i=0}^{n-1} 2i = n(n-1) = 6(5) = 30$. The grass area is $24(18) - 12(8) = 432 - 96 = 336\ \mathrm{m}^2$. Divide by the area cut per minute:

$$t = \frac{336\ \mathrm{m}^2}{30.0\ \mathrm{m}^2/\mathrm{min}} = 11.2\ \mathrm{min}.$$

Practice set based on the supplied Mechatronics slides and homework. Layout modeled on the supplied HW2 problem and solution sheets. This is an independently generated practice exercise.